

SLIDE-BY-SLIDE DELIVERY GUIDE

Computation of Conformal Mappings Presentation

BTP Defense TODAY at 1:00 PM - 60% Marks

SLIDE 1: Title Slide

What to Say:

“Good afternoon professors. Today I’ll present my work on the Computation of Conformal Mappings, specifically focusing on the Schwarz-Christoffel transformation. This is a powerful mathematical tool that bridges pure theory and practical engineering applications.”

Body Language:

- Stand confidently
 - Make eye contact with all professors
 - Pause after greeting
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SLIDE 2: Why Conformal Mapping?

What to Say:

“Let me start with a fundamental question: Why do we need conformal mapping? In engineering and physics, we often face problems with irregular, complex boundaries—think of airflow around an oddly-shaped wing or heat distribution in an irregular metal plate.

*Traditional analytical methods only work in simple geometries like circles or rectangles. Conformal mapping is the **bridge** that transforms these complex domains into simple ones while preserving the governing equations. The diagram on the right shows this visually—we solve the problem in the simple Z-plane and map the solution back.”*

Key Points to Emphasize:

- Use the diagram—point to it while explaining
 - Say “This is **THE** solution to an otherwise intractable problem”
 - Pause after “preserving governing equations”
 - Connect to real-world problems (airflow, heat transfer)
-

SLIDE 3: The Riemann Mapping Theorem

What to Say:

“The theoretical foundation is the Riemann Mapping Theorem, proved by Riemann in 1851 and rigorously established by Schwarz and Hilbert.

*The theorem states: **Any simply connected domain—no matter how irregular—can be conformally mapped to the unit disk.** This is profound because it guarantees existence. However, and this is crucial, it doesn’t tell us **how** to construct the mapping. That’s where Schwarz-Christoffel comes in.”*

Key Points to Emphasize:

- Point to the diagram showing the irregular domain mapping to the disk
 - Emphasize “existence vs. construction”—this sets up the need for SC transformation
 - Make clear the historical significance (1851, later proved rigorously)
 - Connect: “The theorem says it CAN be done, SC shows us HOW”
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SLIDE 4: Foundations of Complex Analysis

What to Say:

*“To understand conformal mappings, we need three key concepts from complex analysis: First, **analytic functions**—functions where the derivative exists everywhere in a region. Second, the **Cauchy-Riemann equations**, which are the necessary and sufficient conditions for analyticity. And third, **harmonic functions**—the real and imaginary parts satisfy Laplace’s equation.*

This connection to Laplace’s equation is why conformal mapping works for problems in electrostatics, fluid flow, and heat transfer—they’re all governed by Laplace’s equation.”

Key Points to Emphasize:

- Don't rush this slide—it's foundational
 - Point to the visual showing orthogonal grids
 - Emphasize the connection: analytic $\rightarrow$ harmonic $\rightarrow$ Laplace's equation
 - If time permits, mention: "The orthogonality shown here is a direct consequence of the Cauchy-Riemann equations"
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SLIDE 5: Conformal Mapping Definition

What to Say:

"What makes a mapping conformal? It must preserve angles—both magnitude and orientation. The sufficient condition is elegant: if $f(z)$ is analytic and its derivative is non-zero, then f is conformal.

Look at the diagram: two curves intersect at angle α in the z -plane. After the conformal map $f(z)$, they still intersect at the same angle α in the w -plane. This angle preservation is what allows physical laws to remain unchanged under the transformation."

Key Points to Emphasize:

- Visual aid: Trace the angle with your hand on the diagram
 - Emphasize the condition: $f'(z) \neq 0$ (local univalence)
 - Connect to physics: "Angle preservation means the form of PDEs is preserved"
 - Mention the key property box on orthogonality if there's time
-

SLIDE 6: Bilinear (Möbius) Transformation

What to Say:

"Before tackling polygons, let's look at the simplest conformal map: the Bilinear or Möbius transformation, $w = \frac{az+b}{cz+d}$.

Its superpower is that it maps circles and lines to circles and lines. This makes it perfect for preliminary transformations—for instance, mapping the upper half-plane to the unit disk. Any Bilinear transformation can be decomposed into translation, inversion, and rotation."

Key Points to Emphasize:

- Keep it brief—this is a stepping stone to SC
 - Point to the diagram showing circle/line preservation
 - Mention the condition: $ad - bc \neq 0$ (ensures invertibility)
 - Connect: “We’ll use this as a building block in more complex mappings”
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SLIDE 7: Schwarz-Christoffel Transformation - Purpose

What to Say:

*“Now we arrive at the main contribution: the Schwarz-Christoffel transformation. It maps the upper half-plane onto the interior of **any** simple polygon. The formula is shown here. The key insight is in the product term: each factor $(z - x_k)^{-\alpha_k}$ controls what happens at a vertex. The points x_k on the real axis map to the polygon vertices, and α_k are the exterior angles.”*

Key Points to Emphasize:

- Slow down here—this is the core
 - Write or point to the formula: $\frac{dw}{dz} = A \prod (z - x_k)^{-\alpha_k}$
 - Explain each component:
 - x_k : pre-images on real axis
 - α_k : exterior angle parameters
 - A, B : normalization constants
 - Emphasize: “This explicit formula is what makes SC so powerful”
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SLIDE 8: Schwarz-Christoffel Derivation Concept

What to Say:

*“How does this formula work? The elegant idea is to control the **argument** (angle) of $\frac{dw}{dz}$ as z moves along the real axis.*

As z crosses a point x_k , the factor $(z - x_k)^{-\alpha_k}$ causes the argument to change by exactly $\alpha_k\pi$ —the exterior angle needed at that vertex. The diagram shows this: as z moves along the real axis in the upper half-plane, w traces the polygon boundary, turning by the correct angle at each vertex.”

Key Points to Emphasize:

- This is the ‘aha’ moment—make it clear!
 - Use the diagram extensively—show the correspondence
 - Explain the argument change: “When z crosses x_k from left to right, $\arg(z - x_k)$ changes by $-\pi$, so $\arg((z - x_k)^{-\alpha_k})$ changes by $\alpha_k\pi$ ”
 - Connect boundary mapping: “Real axis $\rightarrow$ polygon boundary”
 - Emphasize: “This is the genius of Schwarz and Christoffel—encoding geometry in the derivative”
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SLIDE 9: Schwarz-Christoffel Construction Steps

What to Say:

“To construct an SC mapping, we follow four steps:

- 1. Identify where on the real axis each vertex maps to—these are the x_k points*
- 2. Determine the exterior angles from the polygon’s interior angles using $\alpha_k = 1 - \theta_k/\pi$*
- 3. Construct the product ensuring each factor contributes the correct turn*
- 4. Integrate to get w , where A and B control scale, rotation, and position.”*

Key Points to Emphasize:

- Go through each step methodically
 - Point to the diagram showing the UHP and polygon
 - If asked about the integral: “The integral often involves elliptic functions, which is why numerical methods are typically used in practice”
 - Mention the constraint: $\sum \alpha_k = 2$ (polygon closure condition)
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SLIDE 10: Rectangle Mapping Example

What to Say:

*“Let’s see a concrete example: mapping to a rectangle. All four interior angles are $\pi/2$, so all exterior angle parameters $\alpha_k = 1 - (\pi/2)/\pi = 1/2$. The differential becomes $\frac{dw}{dz} = \frac{A}{\sqrt{(z-x_1)(z-x_2)(z-x_3)(z-x_4)}}$. This integral is an **elliptic integral**—it cannot be expressed in elementary functions. This shows that even for a simple rectangle, we need special functions or numerical methods.”*

Key Points to Emphasize:

- Point out the visual: Show how the UHP maps to the rectangle
 - Highlight the four vertices and their pre-images
 - Emphasize: “Even simple polygons lead to complex integrals”
 - If time permits: “The ratio of rectangle sides depends on the ratio of complete elliptic integrals”
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SLIDE 11: Schwarz-Christoffel More Examples

What to Say:

“Slide 11 shows three more examples demonstrating the versatility of SC mappings:
Example 1: *The unit disk—while typically handled by Bilinear transformations, it shows the connection between different conformal maps.*
Example 2: *A semi-infinite strip where the integral simplifies to $\cosh^{-1}(z)$, giving us a closed-form solution.*
Example 3: *A triangular region with specific angles $\pi/4, \pi/4, \pi/2$, showing how different angle combinations affect the formula.”*

Key Points to Emphasize:

- Don’t linger—these are for completeness
 - Mention that some cases yield elementary functions (strip), others don’t (general triangle)
 - If asked: “The semi-infinite strip is important for channel flow problems”
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SLIDE 12: Real-World Applications

What to Say:

*“Why does this matter? Conformal mapping has revolutionized multiple fields: In **fluid dynamics**, the Joukowski transformation maps circles to airfoil shapes, enabling the calculation of lift. In **electrostatics**, it solves for field lines around complex conductor shapes. In **heat transfer**, it analyzes temperature distributions in irregular domains. And in modern **computational engineering**, it generates high-quality grids for numerical simulations.”*

Key Points to Emphasize:

- Be enthusiastic here—show real-world impact!
 - Point to each application icon in the diagram
 - Connect to the central image showing the five applications
 - If time permits, elaborate on one: “In aerodynamics, conformal mapping was crucial before CFD—it gave analytical predictions of lift forces”
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SLIDE 13: Conclusion and Future Directions

What to Say:

“To conclude, conformal mapping—anchored by the Riemann Mapping Theorem and made explicit through the Schwarz-Christoffel formula—provides a powerful bridge between theory and application. It transforms intractable boundary value problems into solvable ones.

Looking forward, future work includes:

- *Validation through simulations and numerical examples*
- *Developing faster numerical algorithms for the parameter problem*
- *Advanced techniques for multiply-connected domains*
- *Applications to modern boundary-value problems in engineering*

The interplay between theory and computation continues to drive innovation in this field. Thank you for your attention. I’m happy to answer questions.”

Key Points to Emphasize:

- Summarize the three key boxes: Theoretical Foundation, Key Transformations, Conformal Mapping

- Highlight the future scope bullet points
 - End with confidence and openness to questions
 - Maintain eye contact during the closing
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GENERAL DELIVERY TIPS

Voice & Pace:

- Speak at 120-130 words per minute (moderate pace)
- Pause after key statements (2-3 seconds)
- Vary your tone—don't be monotone
- Slow down for complex equations

Body Language:

- Stand tall, don't lean on the podium
- Use hand gestures to emphasize points
- Make eye contact with all professors (rotate your gaze)
- Point to diagrams and equations as you discuss them

Engagement Strategies:

- Use the pointer/cursor to guide attention
- Ask rhetorical questions: "Why is this important?"
- Connect slides: "Building on what we just saw..."
- Use transitional phrases: "This leads us to...", "The key insight is..."

Mathematical Rigor:

- Don't skip over equations—explain at least the main SC formula in detail
- If asked to derive something, confidently walk through it
- Be prepared to explain any symbol or term on your slides
- Have alternate explanations ready for complex concepts

SLIDE TRANSITIONS

- **Slide 1 → 2:** “Let me begin by motivating why this topic matters...”
 - **Slide 2 → 3:** “The theoretical foundation that makes all this possible is...”
 - **Slide 3 → 4:** “To understand how this works, we need some complex analysis...”
 - **Slide 4 → 5:** “These concepts lead us to the definition of conformal mapping...”
 - **Slide 5 → 6:** “A simple but powerful example of a conformal map is...”
 - **Slide 6 → 7:** “Now we’re ready for the main result...”
 - **Slide 7 → 8:** “But how does this formula actually work? Let’s derive the concept...”
 - **Slide 8 → 9:** “Let me break down the construction process...”
 - **Slide 9 → 10:** “Let’s make this concrete with an example...”
 - **Slide 10 → 11:** “Here are additional examples showing the range of applications...”
 - **Slide 11 → 12:** “All of this theory translates into powerful real-world applications...”
 - **Slide 12 → 13:** “In conclusion...”
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TIME MANAGEMENT (for ~15-minute presentation)

Slide	Time
Slide 1	1 minute
Slide 2	2 minutes
Slide 3	1.5 minutes
Slide 4	1.5 minutes
Slide 5	1.5 minutes
Slide 6	1 minute
Slide 7	2 minutes (KEY SLIDE)
Slide 8	2 minutes (KEY SLIDE)
Slide 9	1.5 minutes
Slide 10	1.5 minutes
Slide 11	1 minute
Slide 12	1.5 minutes
Slide 13	1.5 minutes
Total	~18 minutes

Note: Adjust based on actual time limit

CONFIDENCE REMINDERS

You know this material deeply—you wrote a comprehensive report on it

The PPT is well-structured and visually clear

Your work connects theory (Riemann) $\rightarrow$ method (SC) $\rightarrow$ applications (real world)

Professors want to see understanding, not perfection

Pauses are powerful—they show confidence

You've got this!

END OF DELIVERY GUIDE

Best of luck with your presentation tomorrow!